

Malaria is caused by the parasite *Plasmodium falciparum* and is transmitted through a female mosquito bite. Although antimalarial treatments are available, *P. falciparum* has become resistant to many of these drugs. *P. falciparum* cells contain a respiratory organelle called the apicoplast that is necessary for the parasite's survival but is not found in humans. Therefore, a new drug that targets this organelle could be useful in the treatment of malaria.

A portion of the apicoplast protein ferredoxin (*Pf*Fd) is shown in Figure 1, with certain amino acid residues labeled. In the apicoplast, ferredoxin NADP⁺ reductase (*Pf*FNR) interacts electrostatically with *Pf*Fd and catalyzes an electron transfer reaction. A compound that selectively inhibits this interaction could be a useful antimalarial agent.

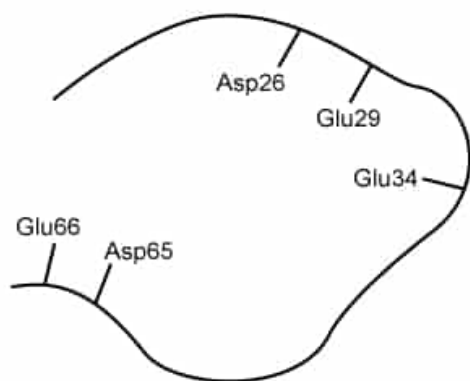
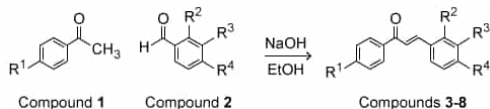


Figure 1 Structure of a portion of *Pf*Fd

Chalcone (Compound **3**) is a compound that interacts with *Pf*Fd and exhibits antimalarial properties. A series of chalcone derivatives (Compounds **4–8**) were synthesized via the aldol condensation shown in Scheme 1, where nucleophilic addition of Compound **1** to Compound **2** is followed by elimination to yield α,β -unsaturated carbonyl molecules, Compounds **3–8**.



Scheme 1

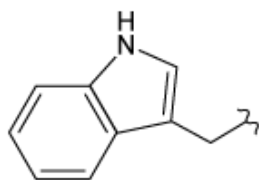
An inhibition assay was performed to compare chalcone (Compound **3**) and its derivatives by determining the extent to which these compounds inhibited electron transfer between *Pf*FNR and *Pf*Fd (Table 1).

Table 1 Inhibition of Electron Transfer Assay Results

Compound	R ¹	R ²	R ³	R ⁴	Inhibition (%)
3	H	H	H	H	23.06
4	NH ₂	H	H	H	11.94
5	H	H	H	OCH ₃	5.69
6	H	H	OCH ₃	H	1.00
7	H	OCH ₃	H	H	-9.29
8	NH ₂	H	H	OCH ₃	51.00

Which amino acid residue on *Pf*FNR most likely interacts with the residues of *Pf*Fd shown in Figure 1?

☐ A.



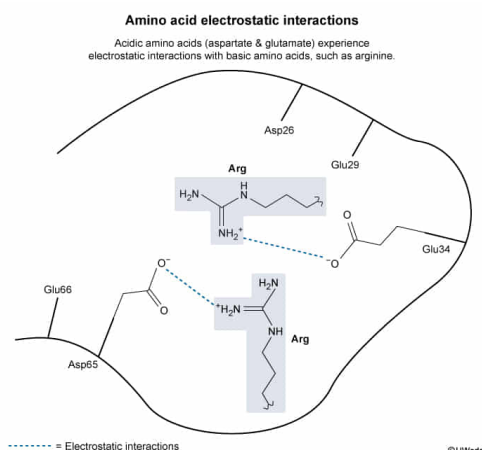
[3%]

- ☐ B. [O-]C(=O)CCC [12%]
- ☒ C. NC(=N)CCCC[NH2+] [79%]
- ☐ D. [O-]C(=O)CC [5%]

Correct

79% answered correctly

Explanation:



Amino acids have three components: an amino group, a carboxyl group, and a **side chain**, all of which are bonded to the same carbon atom, known as the α -carbon. Amino acid structures differ only by the makeup of the substituent side chains, which contain different **functional groups** with specific **characteristics**. Therefore, amino acids can be **classified** based on their side-chain characteristics. Some amino acids fall into **more than one** category (eg, both hydrophilic and basic).

Acidic and basic amino acids have negatively and positively charged side chains, respectively. These charged amino acids can experience **electrostatic interactions** with each other or with other oppositely charged functional groups.

Figure 1 shows that *PfF*d contains the acidic amino acid residues aspartate (Asp, D) and glutamate (Glu, E). The passage states that *PfF*d and *PfF*NR experience electrostatic interactions, indicating that the amino acid residues have a positive-negative charge attraction. Therefore, a positively charged basic amino acid residue on *PfF*NR will most likely interact with the negatively charged acidic residues in *PfF*d. The only structure shown with a positively charged side chain is **arginine**.

(Choice A) This structure is tryptophan (Trp, W), a hydrophobic amino acid with an uncharged side chain. Because its side chain is uncharged, tryptophan cannot interact with the acidic amino acid residues of *PfF*d.

(Choices B and D) These structures are glutamate (Glu, E) and aspartate (Asp, D), which are acidic amino acids with negatively charged side chains. They will not interact with the negatively charged Glu and Asp residues shown in Figure 1 but will instead repel each other.

Educational objective:

Amino acids have three components: an amino group, a carboxyl group, and a side chain. The side chains are composed of different functional groups with varying characteristics, allowing the amino acids to be categorized by these characteristics. Positively charged side chains can interact with negatively charged ones, and vice versa.

Time spent:0 sec

Subject:
Organic Chemistry

QID:402303

System:
Functional Groups and Their Reactions

Malaria is caused by the parasite *Plasmodium falciparum* and is transmitted through a female mosquito bite. Although antimalarial treatments are available, *P. falciparum* has become resistant to many of these drugs. *P. falciparum* cells contain a respiratory organelle called the apicoplast that is necessary for the parasite's survival but is not found in humans. Therefore, a new drug that targets this organelle could be useful in the treatment of malaria.

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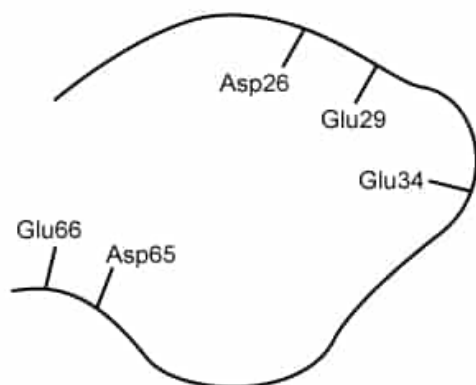
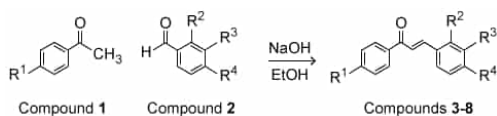


Figure 1 Structure of a portion of *PfFd*

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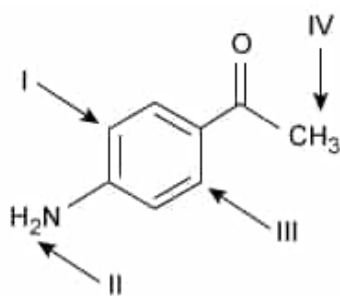
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The aldol condensation is an important carbon–carbon bond-forming reaction that requires a compound with an α -H.



Given that an enolate is formed in the first step of the aldol condensation, which site will be deprotonated by a base?

- ☐ A. I [2%]
☐ B. II [13%]
☐ C. III [3%]
✓ ☒ D. IV [80%]

Correct

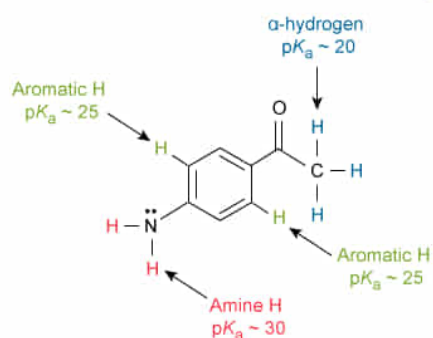
79% answered correctly

Explanation:

Hydrogen atom acidity

Amine hydrogen < aromatic hydrogen < α -hydrogen

Increasing acidity



Because α -hydrogen atoms are the most acidic in Compound 1, a base will deprotonate it, forming a resonance-stabilized enolate.

Carbon atoms adjacent to a carbonyl are known as α -carbons. The oxygen on the carbonyl carbon is electron withdrawing, which gives the carbonyl carbon a partial positive charge and results in a decreased electron density around the α -protons. For this reason, protons on the α -carbon are more acidic than other protons bonded to a carbon atom and have much **lower** pK_a values. Therefore, a **base** can more **easily deprotonate** an **α -carbon** than other carbons.

In addition, the negative charge resulting from deprotonation can be delocalized to the oxygen of the carbonyl, forming an **enolate**. Because oxygen is more **electronegative** than carbon, the enolate is more stable.

The labeled protons on the carbon adjacent to the carbonyl (the α -carbon) are acidic and have a lower pK_a compared to other protons in the compound. Therefore, these α -protons can be removed by a base to form an enolate by **delocalization** of the negative charge.

(Choices A and C) These are aromatic hydrogen atoms and will not likely be removed by a base, as the resulting carbanion is not as stable because the negative charge cannot be delocalized to the carbonyl oxygen.

(Choice B) Amines have high pK_a values, indicating that this functional group is **basic** and more likely to be protonated than deprotonated.

Educational objective:

Protons on an α -carbon (ie, adjacent to a carbonyl) are more acidic than other protons bonded to a carbon atom because the carbonyl oxygen is electron withdrawing, resulting in less electron density around the α -protons. Therefore, α -protons have lower pK_a values and can be more easily removed by a base to form an enolate, which can be stabilized by charge delocalization.

Time spent: 0 sec

Subject:
Organic Chemistry

QID: 402304

System:
Functional Groups and Their Reactions

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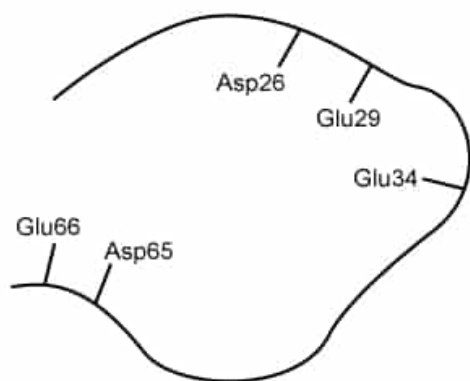
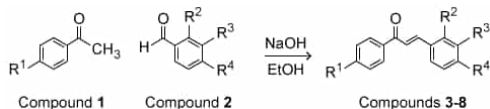


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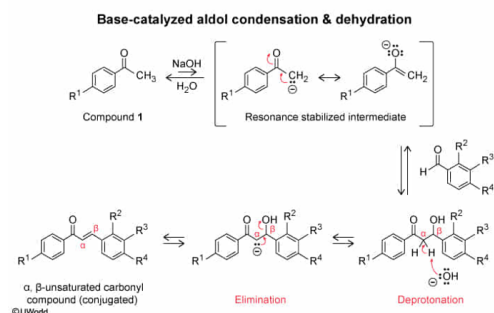
Which statement does NOT describe the aldol condensation explained in the passage?

- ☐ A. The reaction has two carbonyl compounds as substrates. [6%]
- ☒ B. The reaction involves protonation of a carbon for elimination. [76%]
- ☐ C. The reaction has a resonance-stabilized intermediate. [11%]
- ☐ D. The reaction yields a conjugated product. [5%]

Correct

76% answered correctly

Explanation:



Aldol condensations are important carbon-carbon bond-forming reactions where two carbonyl compounds (ketones and/or aldehydes) are joined together by nucleophilic addition to form the **aldol product** (ie, a β -hydroxy ketone). The carbonyls can come from separate compounds (intermolecular aldol condensation) or from a dicarbonyl compound (**intramolecular aldol condensation**). The aldol product can then undergo an elimination to form an α, β -unsaturated carbonyl compound.

An aldol condensation as described in the passage requires **two carbonyl substrates** (Compounds **1** and **2**) (**Choice A**). The α -carbon (the carbon adjacent to the carbonyl) on Compound **1** is deprotonated when a base is added. The **intermediate** formed is **stabilized** through **resonance** as the negative charge on the α -carbon is delocalized to the carbonyl oxygen, forming an enolate (**Choice C**). The nucleophilic enolate (from Compound **1**) then attacks the carbonyl of Compound **2**. A new carbon-carbon bond is formed (the aldol product), in which the carbonyl adjacent to the nucleophile remains intact. Finally, $-\text{OH}$ elimination takes place to give a **conjugated** product, an **α, β -unsaturated enone** (**Choice D**).

The **elimination** step requires the **α -carbon** to be **deprotonated** rather than protonated. The resulting carbanion is an enolate that can eliminate the $-\text{OH}$ on the β -carbon and form a double bond between the α - and β -carbons, giving a conjugated product.

Therefore, **no protonation steps** exist in the reaction described in the passage.

Educational objective:

Aldol condensations are carbon-carbon bond-forming reactions that require two carbonyl substrates (ketones and/or aldehydes). The reaction begins with deprotonation of the α -carbon on one of the substrates, forming a resonance-stabilized enolate intermediate. Nucleophilic addition of the enolate to

the other carbonyl substrate yields the aldol product; deprotonation of the α -carbon, followed by $-\text{OH}$ elimination, then yields the conjugated product, an α, β -unsaturated carbonyl compound.

Time spent: 0 sec

Subject:
Organic Chemistry

QID: 402305

System:
Functional Groups and Their Reactions

Malaria is caused by the parasite *Plasmodium falciparum* and is transmitted through a female mosquito bite. Although antimalarial treatments are available, *P. falciparum* has become resistant to many of these drugs. *P. falciparum* cells contain a respiratory organelle called the apicoplast that is necessary for the parasite's survival but is not found in humans. Therefore, a new drug that targets this organelle could be useful in the treatment of malaria.

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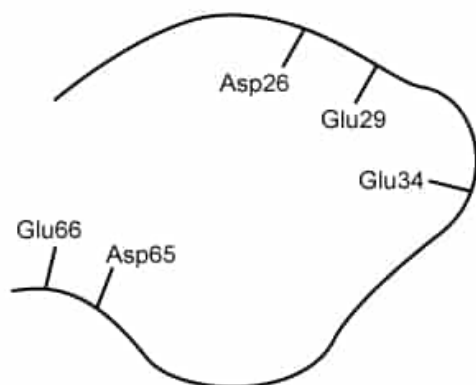
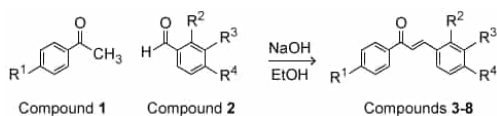


Figure 1 Structure of a portion of *PfFd*

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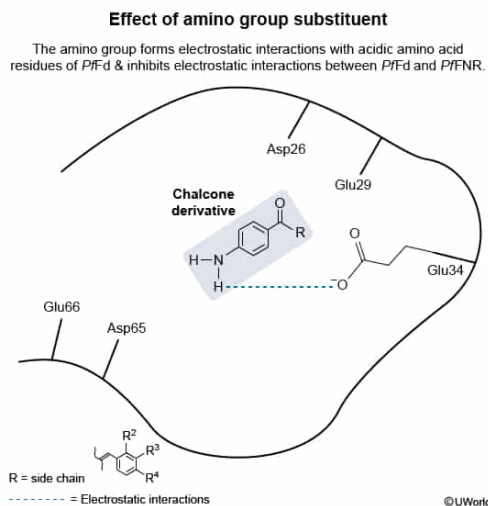
What is the most likely explanation why an amino group was chosen as a substituent on the chalcone derivatives?

- ☐ A. The amino group increases the acidity of the chalcone derivatives. [9%]
- ☐ B. The amino group improves the stability of the chalcone derivatives. [28%]
- ✓ ☒ C. The amino group interferes with the *Pf*Fd–*Pf*FNR electrostatic interaction. [59%]
- ☐ D. The amino group makes the chalcone derivative more nonpolar than chalcone. [2%]

Correct

58% answered correctly

Explanation:



Basic and acidic functional groups can experience **electrostatic interactions** due to the positive–negative charge attraction. The passage states that the protein ***Pf*Fd has an electrostatic interaction with *Pf*FNR** in the apicoplast, which is required for the parasite's (*P. falciparum*) survival. Figure 1 shows negatively charged **acidic amino acid** residues on *Pf*Fd, so positively charged **basic amino acid** residues of *Pf*FNR most likely interact with the acidic residues in *Pf*Fd.

According to the passage, chalcone (Compound **3**) exhibits antimalarial properties. Chalcone derivatives (Compounds **4–8**) were synthesized in an effort to create a more active antimalarial compound. The **amine group** substituent consists of an electronegative nitrogen bonded to hydrogens. Due to the difference in electronegativity, the hydrogen holds a partial positive charge. This partial positive charge likely allows the chalcone derivative to **interact** with the **acidic residues** of *Pf*Fd,

blocking the basic amino acid residues of *Pf*FNR from interacting. Therefore, the amino group was most likely chosen to **interfere with the *Pf*Fd–*Pf*FNR electrostatic interaction**.

(Choice A) An amine is a basic functional group, and therefore would increase the *basicity* rather than the acidity of the chalcone derivatives.

(Choice B) Although amino groups attached to aromatic rings are typically stable, the passage does not indicate whether the substituents on the chalcone derivatives affected the compound's overall stability.

(Choice D) The lone pair of electrons on nitrogen creates a large **dipole**, pulling electrons toward the nitrogen and away from the carbon. Therefore, the amino group *increases* the **polarity** of the compound.

Educational objective:

Compounds with basic and acidic functional groups, such as amino acids with basic and acidic side chains, can form electrostatic interactions with each other due to positive–negative charge attraction of the respective functional groups. These interactions can be disrupted by other charged molecules that

compete for binding.

Time spent:0 sec

Subject:
Organic Chemistry

QID:402306

System:
Functional Groups and Their Reactions

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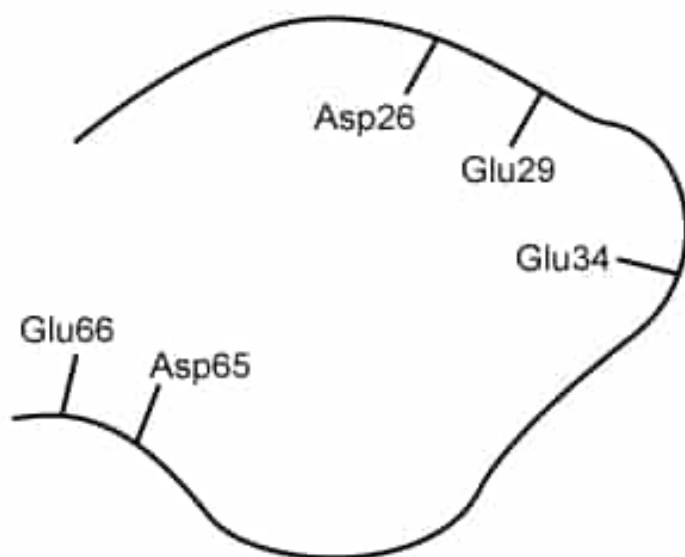
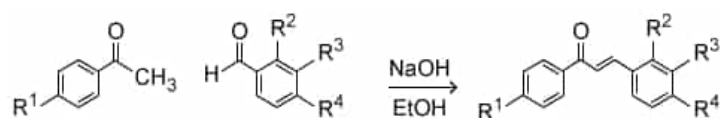


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7	H	OCH ₃	H	H	-9.29
8	NH ₂	H	H	OCH ₃	51.00

Based on the data in Table 1, what effect do the chalcone derivative substituents have on percent inhibition, relative to chalcone?

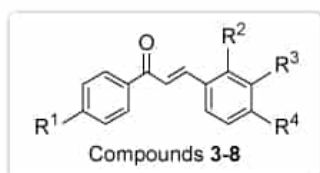
- ☐ A. An NH₂ substituent alone has a positive impact on percent inhibition, but an OCH₃ substituent alone has no impact. [1%]
- ☐ B. An NH₂ substituent alone has a positive impact on percent inhibition, but an OCH₃ substituent alone has a negative impact. [7%]
- ✓ ☒ C. Chalcone derivatives with NH₂ and OCH₃ substituents impact the percent inhibition and are both necessary to increase the percent inhibition. [79%]
- ☐ D. Chalcone derivatives with NH₂ and OCH₃ substituents impact the percent inhibition, but only one is necessary to increase the percent inhibition. [10%]

Correct

80% answered correctly

Explanation:

Effect of chalcone derivative substituents on percent inhibition



	Compound	R ¹	R ²	R ³	R ⁴	Inhibition (%)
Chalcone	3	H	H	H	H	23.06
Chalcone derivative	4	NH ₂	H	H	H	11.94
	5	H	H	H	OCH ₃	5.69
	6	H	H	OCH ₃	H	1.00
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■ = higher percent inhibition than chalcone
■ = lower percent inhibition than chalcone

NH₂ & OCH₃ substituents impact the percent inhibition & are both necessary to increase the percent inhibition.

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The chalcone derivatives were synthesized in an effort to improve the antimalarial activity of chalcone. The passage states a compound that could selectively inhibit the interaction between *PfFNR* and *PfFd* could be a beneficial antimalarial agent. Chalcone (Compound **3**) and several derivatives (Compounds **4–8**) were tested in an assay to determine the effect the compounds had on the electron transfer between *PfFNR* and *PfFd*. Table 1 indicates that **chalcone inhibits 23.06%** of the interaction between *PfFNR* and *PfFd*.

Each **chalcone derivative** has a different **impact** on the electron transfer reaction. Table 1 shows that chalcone (Compound **3**) has a H atom for every R group. The derivatives (Compounds **4–7**) each have only one substituent (either NH₂ or OCH₃), and these compounds are *weaker inhibitors* than chalcone because their percent inhibition values are *lower than* 23.06%. Compound **8** has *both* NH₂ and OCH₃ substituents. The percent inhibition for this derivative is **51%**, indicating that Compound **8** inhibits the interaction between *PfFNR* and *PfFd* to a **greater extent than does chalcone**. Therefore, **both the NH₂ and OCH₃ substituents are necessary to increase the percent inhibition relative to chalcone**.

(Choices A, B, and D) The chalcone derivative substituents each have an impact on the percent inhibition *compared to chalcone*. Alone, NH_2 and OCH_3 each have a *negative* effect because the percent inhibition for Compounds **4–7** is lower than that of chalcone (23.06%). The data in Table 1 indicate that *both* substituents are needed to increase the percent inhibition *relative to chalcone*.

Educational objective:

Structural modifications to biologically active molecules are performed to determine whether the biological activity of the molecule can be increased. To determine the effect of the structural modifications, the activity of each modified molecule should be compared to that of the original molecule.

Time spent:0 sec

Subject:
Organic Chemistry

QID:402307

System:
Functional Groups and Their Reactions

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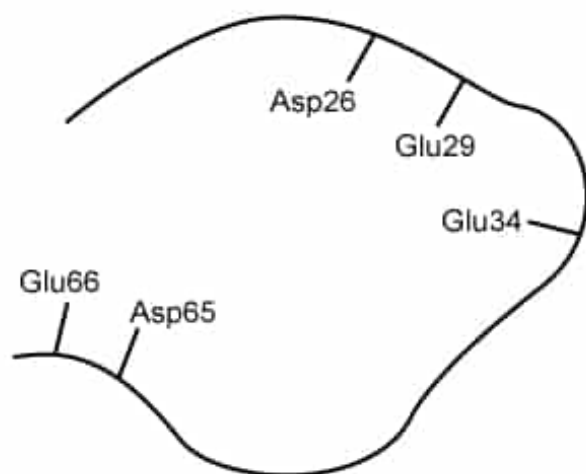
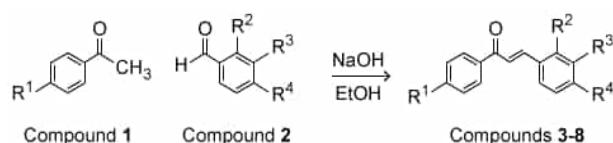


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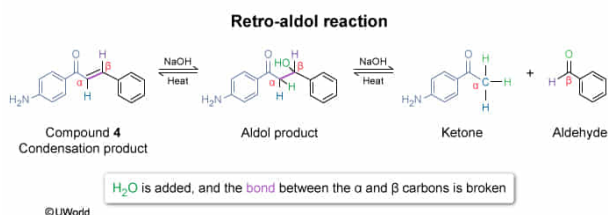
If Compound **4** is heated in an aqueous base, which of the following functional groups would result from this reaction?

- ✓ ☒ A. A ketone and an aldehyde [42%]
- ☐ B. A ketone and an alcohol [40%]
- ☐ C. Two aldehydes [8%]
- ☐ D. Two ketones [8%]

Correct

43% answered correctly

Explanation:



A **retro-aldol** reaction is the reverse of an aldol condensation. When **heated** and treated with an aqueous **base**, the dehydration product (**enone** or **enal**) is first hydrated into the aldol product (**β -hydroxy ketone**), and then the **carbon-carbon bond** between the α - and β -carbons **breaks**. The retro-aldol reaction yields **two products**: either two ketones, two aldehydes, or one of each.

The **products** **depend** on the **substituents** on the **carbonyl carbon** and the **β -carbon**. If the carbonyl carbon is bonded to a hydrogen atom before the reaction (ie, it was already an aldehyde), it will still be an aldehyde after the reaction. If the β -carbon is bonded to a hydrogen, it will become an aldehyde. If the carbonyl or β -carbon are not bonded to hydrogen, they will become ketones.

Compound **4** is an enone in which **the β -carbon is bonded to hydrogen** and the **carbonyl carbon is not**. Therefore, a **ketone** will form from the carbonyl and an **aldehyde** will form from the β -carbon when Compound **4** is heated and treated with

an aqueous base.

(Choice B) A ketone is formed in this retro-aldol reaction, but the other product is an *aldehyde* rather than an alcohol. An alcohol cannot be a product of a retro-aldol reaction.

(Choice C) Because the carbonyl carbon is not bonded to any hydrogen atoms, it will not form an aldehyde.

(Choice D) Because the β -carbon contains a hydrogen substituent, it will not form a ketone.

Educational objective:

The retro-aldol reaction is the reverse of the aldol condensation. When heated and treated with an aqueous base, the carbon-carbon bond between the α - and β -carbons is broken, forming two ketones, two aldehydes, or one of each, depending on the substituents on the carbonyl carbon and the β -carbon. Aldehydes form if the carbonyl and/or β -carbon has hydrogen substituents whereas ketones form if all substituents are carbon chains.

Time spent:0 sec

Subject:

Organic Chemistry

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System:

Functional Groups and Their Reactions